KESTREL RIDGE TECHNOLOGIES LTD. - CALCULATION SHEET CS-042 (FICTIONAL)

Shrink-fit coupling for hub H-42

8. Interference fit data

(1) Shaft diameter, 180 mm

(2) Hub outer diameter, 320 mm

(3) Contact length, 210 mm

9. Our proposal of push-up distance, (Pd), mm

(1) Push-up distance, 10.6 mm

(2) Grip stress corresponding to Pd, N/mm² $\sigma ra = Pd / Pr \times \sigma r = 111$ N/mm²

(3) Safety factor against slip, $S = \mu \times \sigma ra \times A / F = 2.4$

10. Variables

A : Interfacial surface area ..... A= 1,118,355

Pr : Reference push-up distance, mm ..... Pr = 8.2

σr : Grip stress at Pr, N/mm² ..... $\sigma r = 85.8$

μ : Friction coefficient ..... $\mu = 0.14$

Θ : Taper angle, deg ..... $\Theta = 1.5$

11. Hydraulic oil injection

(1) Required oil pressure, 142 N/mm²

(2) Maximum pump pressure, 200 N/mm²

12. Remarks

The push-up distance shall be checked with a dial gauge at 20 °C.

Checked by: J. Doe (fictional)